

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the series $\sum_{n=2}^{\infty} ne^n$.

- a) Explain how we can know that this series diverges?

Very clearly we see that $\lim_{n \rightarrow \infty} ne^n = \infty \neq 0$ and therefore, by the Test for Divergence, the series must diverge.

- b) If this series is changed to $\sum_{n=2}^{\infty} ne^{-n}$ it now converges. Find a way to prove this series converges.

Let $f(x) = xe^{-x} = \frac{x}{e^x}$. Note that $f'(x) = \frac{e^x - xe^x}{e^{2x}} = \frac{1-x}{e^x}$ which is negative for all values of $x \geq 2$.

This means that $f(x)$ is decreasing on $[2, \infty]$. We also note that $f(x)$ is positive and continuous on $[2, \infty]$.

We then calculate that $\int_2^{\infty} xe^{-x} dx = \lim_{t \rightarrow \infty} \left[\int_2^t xe^{-x} dx \right]$.

Using IBP let $u = x$ & $dv = e^{-x} dx$ and thus $du = dx$ & $v = -e^{-x}$. Therefore we have that

$$\begin{aligned} \int_2^{\infty} xe^{-x} dx &= \lim_{t \rightarrow \infty} \left[\int_2^t xe^{-x} dx \right] = \lim_{t \rightarrow \infty} \left[-xe^{-x} \Big|_2^t + \int_2^t e^{-x} dx \right] = \lim_{t \rightarrow \infty} \left[\left(-te^{-t} + 2e^{-2} \right) - e^{-x} \Big|_2^t \right] = \lim_{t \rightarrow \infty} \left[\left(-\frac{t}{e^t} + 2e^{-2} \right) - \left(e^{-t} - e^{-2} \right) \right] \\ &= \lim_{t \rightarrow \infty} \left[\left(-\frac{t}{e^t} - \frac{1}{e^t} \right) + (2e^{-2} + e^{-2}) \right] = \lim_{t \rightarrow \infty} \left[\left(-\frac{t}{e^t} - \frac{1}{e^t} \right) \right] + (2e^{-2} + e^{-2}) \\ &\stackrel{H}{=} \lim_{t \rightarrow \infty} \left[\left(-\frac{1}{e^t} - \frac{1}{e^t} \right) \right] + (2e^{-2} + e^{-2}) \\ &= 2e^{-2} + e^{-2} \end{aligned}$$

Thus, $\int_2^{\infty} xe^{-x} dx$ converges and therefore, by the Integral Test, the series $\sum_{n=2}^{\infty} ne^{-n}$ converges.

2. Use any of the techniques discussed in class thus far to determine if the following series converge or diverge. If possible, determine what the series converges to.

a) $\sum_{k=1}^{\infty} \frac{3^{k+1}}{6^{2k}}$

We first rewrite the series $\sum_{k=1}^{\infty} \frac{3^{k+1}}{6^{2k}} = \sum_{k=1}^{\infty} 3 \frac{3^k}{36^k} = \sum_{k=1}^{\infty} \frac{1}{4} \left(\frac{1}{12} \right)^{k-1}$. This is a geometric series with $a = \frac{1}{4}$ and $r = \frac{1}{12}$. We see that $|r| < 1$ and so the series converges to $\frac{a}{1-r} = \frac{1/4}{1-1/12} = \frac{1/4}{11/12} = \frac{12}{44} = \frac{3}{11}$.

b) $\sum_{k=1}^{\infty} \frac{k+1}{k^{3/2} + \sqrt{k}}$

Note that $\sum_{k=1}^{\infty} \frac{k+1}{k^{3/2} + \sqrt{k}} = \sum_{k=1}^{\infty} \frac{k+1}{\sqrt{k}(k+1)} = \sum_{k=1}^{\infty} \frac{1}{k^{1/2}}$. This is a p-series with $p = \frac{1}{2}$. Since $p = \frac{1}{2} < 1$, this series diverges.

c) $\sum_{k=2}^{\infty} \frac{e^k}{\ln(k)}$

Note that $\lim_{k \rightarrow \infty} \frac{e^k}{\ln(k)} \stackrel{H}{=} \lim_{k \rightarrow \infty} \frac{e^k}{1/k} = \lim_{k \rightarrow \infty} k e^k = \infty \neq 0$. Therefore, by the Test for Divergence, the series diverges.

d) $\sum_{k=1}^{\infty} \left(\frac{3^{k+1}}{5^k} + \frac{k^2}{k^4} \right)$

Note that $\sum_{k=1}^{\infty} \left(\frac{3^{k+1}}{5^k} + \frac{k^2}{k^4} \right) = \sum_{k=1}^{\infty} \left(\frac{9}{5} \left(\frac{3}{5} \right)^{k-1} + \frac{1}{k^2} \right) = \sum_{k=1}^{\infty} \frac{9}{5} \left(\frac{3}{5} \right)^{k-1} + \sum_{k=1}^{\infty} \frac{1}{k^2}$. The first series is a convergent geometric series since $r = \frac{3}{5}$ and $|r| < 1$. The second series is a p-series with $p = 2$ and since $p = 2 > 1$ the series converges.

e) $\sum_{k=1}^{\infty} \sin^2(k) \cos(k)$

We see that $\lim_{k \rightarrow \infty} \sin^2(k) \cos(k)$ DNE and so, by the Test for Divergence, the series diverges.

f) $\sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1} \right)$

We suspect that this is a telescoping series. We start by finding the nth partial sum of the series.

$$s_n = \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{n+1}.$$

Note that $\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right) = 1 - 0 = 1$.

Therefore, the series converges to 1 by definition.

3. Consider the series $\sum_{n=0}^{\infty} \frac{7n}{n^2+5}$. Suppose we wish to apply the Integral Test to this series.

a) Show conditions for the Integral Test work from the 3rd term onward.

$$\text{Let } f(x) = \frac{7x}{x^2+5}.$$

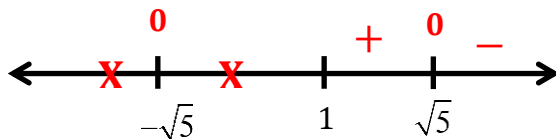
Clearly $f(x)$ is continuous and positive on $[3, \infty)$. Now we check to see if $f(x)$ is decreasing.

$$\text{We see that } f'(x) = \frac{(x^2+5)(7) - 7x(2x)}{(x^2+5)^2} = \frac{35-7x^2}{(x^2+5)^2} \rightarrow x = \sqrt{\frac{35}{7}} = \sqrt{5}.$$

$$f'(x) = 0 \text{ when } 35 - 7x^2 = 0 \rightarrow x = \pm\sqrt{5}.$$

$f'(x)$ is never undefined.

Therefore $f(x)$ has critical values of $\pm\sqrt{5}$. We only care about values past $x=1$ and so we put $x = \sqrt{5}$ on a number line and check if $f(x)$ is increasing or decreasing on either side of it.



We see that after $x = \sqrt{5}$ the function is decreasing.

Therefore, on $[3, \infty)$ $f(x)$ is decreasing and thus $a_n = \frac{7n}{n^2+5}$ is decreasing on $[3, \infty)$.

b) Show behavior of series from 3rd term onward.

The Integral Test applies to the series $\sum_{n=3}^{\infty} \frac{7n}{n^2+5}$.

We then look at the integral $\int_3^{\infty} \frac{7x}{x^2+5} dx$ and we see that if we let $u = x^2+5$ then

$$du = 2x dx \rightarrow \frac{1}{2} du = x dx \text{ and}$$

$$\begin{aligned} \int_3^{\infty} \frac{7x}{x^2+5} dx &= \lim_{t \rightarrow \infty} \left[\int_3^t \frac{7x}{x^2+5} dx \right] = \lim_{t \rightarrow \infty} \left[\int_{14}^{t^2+5} \frac{7}{2} \cdot \frac{1}{u} du \right] = \lim_{t \rightarrow \infty} \left[\frac{7}{2} \ln |u| \Big|_{14}^{t^2+5} \right] = \lim_{t \rightarrow \infty} \left[\frac{7}{2} \ln |t^2+5| - \frac{7}{2} \ln |14| \right] \\ &= \infty - \frac{7}{2} \ln(14) \\ &= \infty \end{aligned}$$

Since the integral diverges the series $\sum_{n=3}^{\infty} \frac{7n}{n^2+5}$ diverges.

c) Make conclusion about full series.

Since we know that $\sum_{n=3}^{\infty} \frac{7n}{n^2+5}$ diverges, the series $\sum_{n=0}^{\infty} \frac{7n}{n^2+5}$ must also diverge since it only adds finitely many extra terms.

4. Find all values of c for which the following series converges $\sum_{n=1}^{\infty} \left(\frac{c}{n} - \frac{1}{n+1} \right)$.

We look at the k th partial sum of the series. $s_k = \left(\frac{c}{1} - \frac{1}{2} \right) + \left(\frac{c}{2} - \frac{1}{3} \right) + \left(\frac{c}{3} - \frac{1}{4} \right) + \dots + \left(\frac{c}{k} - \frac{1}{k+1} \right)$

We then attempt to rewrite this sum. Note that,

$$\begin{aligned} s_k &= \left(\frac{c}{1} - \frac{1}{2} \right) + \left(\frac{c}{2} - \frac{1}{3} \right) + \left(\frac{c}{3} - \frac{1}{4} \right) + \dots + \left(\frac{c}{k} - \frac{1}{k+1} \right) \\ &= \frac{c}{1} + \frac{c-1}{2} + \frac{c-1}{3} + \frac{c-1}{4} + \dots + \frac{c-1}{k} - \frac{1}{k+1} \\ &= c + (c-1) \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{k} \right) - \frac{1}{k+1} \\ &= c + (c-1) \sum_{n=2}^k \frac{1}{n} - \frac{1}{k+1} \end{aligned}$$

We then see that the sum of the series is given by the limit of this partial sum.

$$\text{So } \sum_{n=1}^{\infty} \left(\frac{c}{n} - \frac{1}{n+1} \right) = \lim_{k \rightarrow \infty} s_k = \lim_{k \rightarrow \infty} \left[c + (c-1) \sum_{n=2}^k \frac{1}{n} - \frac{1}{k+1} \right] = c + (c-1) \sum_{n=2}^{\infty} \frac{1}{n}.$$

Since $\sum_{n=2}^{\infty} \frac{1}{n}$ is divergent since it is the Harmonic series with just a single term removed. Therefore, if the result of this limit is to be a finite constant value it must be the case that $c=1$ and thus

$$\sum_{n=1}^{\infty} \left(\frac{c}{n} - \frac{1}{n+1} \right) = \lim_{k \rightarrow \infty} s_k = 1 + (1-1) \sum_{n=2}^{\infty} \frac{1}{n} = 1 - 0 = 1.$$